# The World Health Organization skin neglected tropical diseases app: A dynamic capacity-building training tool enhanced with artificial intelligence

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Skin neglected tropical diseases (NTDs) remain a major public health challenge in low- and middle-income countries, where frontline health workers often lack dermatological training. In response, the World Health Organization created the Skin NTDs App—developed by UniversalDoctor—to support frontline health workers in resource-limited settings. Initially created as a digital adaptation of a World Health Organization training guide, the app evolved by incorporating another clinical decision support tool from Until No Leprosy Remains and an artificial intelligence-powered visual classifier. Our purpose is to describe the World Health Organization Skin NTDs App and evaluate its artificial intelligence-powered visual classifier. The visual classifier was trained to identify 12 out of 13 skin NTDs through a convolutional neural network (DenseNet-121). Performance was assessed through sensitivity, specificity, accuracy, precision, F1-score, and per-class sensitivity from top-1 to top-5. The visual classifier demonstrated high performance in internal evaluations, achieving 99.8% top-5 sensitivity across all diseases and top-1 accuracy above 75% for most diseases. Some underrepresented conditions (eg, chromoblastomycosis, sporotrichosis) showed lower precision. In conclusion, the artificial intelligence-powered World Health Organization Skin NTDs App is a promising digital tool for capacity-building of frontline health workers in underserved areas. Continued development, external validation, and integration into clinical workflows will be critical to assessing its performance globally.

**Keywords:** Artificial intelligence, Clinical decision support tool, Frontline health workers, Mobile health, Skin neglected tropical diseases

## INTRODUCTION

Neglected tropical diseases (NTDs) are a diverse group of 25 conditions or groups of conditions caused by a variety of pathogens (including viruses, bacteria, parasites, fungi, and toxins). They are associated with devastating health, social, and economic consequences. It is estimated that NTDs affect more than 1 billion people, particularly in low- and middle-income countries (World Health Organization, 2025). Among these diseases, skin NTDs constitute a substantial subgroup with at least 13 conditions, characterized by skin lesions that necessitate considerable diagnostic and treatment resources, including trained personnel and financial support (Chandler and Fuller, 2018). Recognizing the pressing need for effective tools in resource-constrained settings, World Health Organization (WHO) created the Skin NTDs App, a digital health solution developed by UniversalDoctor based on the WHO training guide for front-line health workers (FHWs) (World Health Organization, 2018). Usually, 90% or more of the skin conditions found at the primary health level are non-skin NTDs (Ruiz-Postigo et al, 2025;

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*Abbreviations: AI, artificial intelligence; CDST, clinical decision support tool; CNNs, convolutional neural networks; FHWs, front-line health workers; NTDs, neglected tropical diseases; VC, visual classifier; WHO, World Health Organization*

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Yotsu et al, 2018). To address that situation, WHO included 24 common skin conditions in its Skin NTDs App.

Nevertheless, most professionals working at the primary healthcare level do not receive specific training in dermatology. In this sense, the rapid evolution of artificial intelligence (AI) in medical education may play a key role in the differential diagnosis and treatment of skin NTDs. Various AI technologies, such as machine learning, convolutional neural networks (CNNs), support vector machines, decision trees, classification and regression trees, and artificial neural networks, have been introduced for NTD diagnosis (Thunga et al, 2024), focusing on two key areas: (a) image-based diagnosis and (b) diagnosis based on symptomatology and laboratory parameters. AI-based supporting diagnosis tools in dermatology have moved in the last seven years beyond skin cancer alone to a wide range of common skin diseases (Venkatesh et al, 2024).

In the field of NTDs, the application of AI as a clinical decision support tool (CDST) for the identification of skin NTDs has the potential to significantly contribute to building capacity for FHWs. In this sense, the Skin NTDs App has evolved considerably since its inception. The first version, released in 2020, was a digital adaptation of the WHO training guide, which contained a CDST for differential diagnosis of skin NTDs and selected common skin diseases based on four signs, namely lumps, ulcers, swelling, and patches (World Health Organization, 2018). Subsequently, the app underwent further enhancements after being merged with the tool SkinApp developed by until No Leprosy Remains (Mwageni et al, 2024; World Health Organization, 2023b). This allowed the WHO Skin NTDs App to have information on 24 common skin diseases and another CDST.

The app has five sections (Figure 1):

1) Signs and symptoms: it is a CDST that allows users to select a part of the body alongside several signs or symptoms. The result shows a list of possible diseases that exhibit those clinical features.
2) Diagnoses: it contains information about 24 common skin diseases and nine skin NTDs. Clicking on each disease displays images, guidance on making the diagnosis and managing the case, alongside the follow-up.
3) How to manage: it has brief information on how to manage dry scaly skin conditions and wounds/broken skin.
4) Glossary: it is a list of 28 signs with their brief definition.
5) Skin NTDs learning hub: This section corresponds to the first version of the app before it merged with the No Leprosy Remains's SkinApp. It contains four sub-sections (Figure 2):

5.1. Skin NTDs: it has information on 13 diseases. Clicking on each disease displays its description, images, and a map showing the geographical distribution if available.
5.2. Global Index: this section lists all countries. By selecting a country, one can know which skin NTDs are endemic in that location, alongside the latest data on the number of cases.
5.3. Diagnosis: this section contains information about the four skin signs used in the second CDST, that is, lumps, ulcers, swelling, and patches. The CDST allows users to select one or more countries where the patient may have been infected, and to select one or more of the main signs and other signs or symptoms, which will lead to a list of possible causes.
5.4. Information: this section contains information about the skin and links for further information.

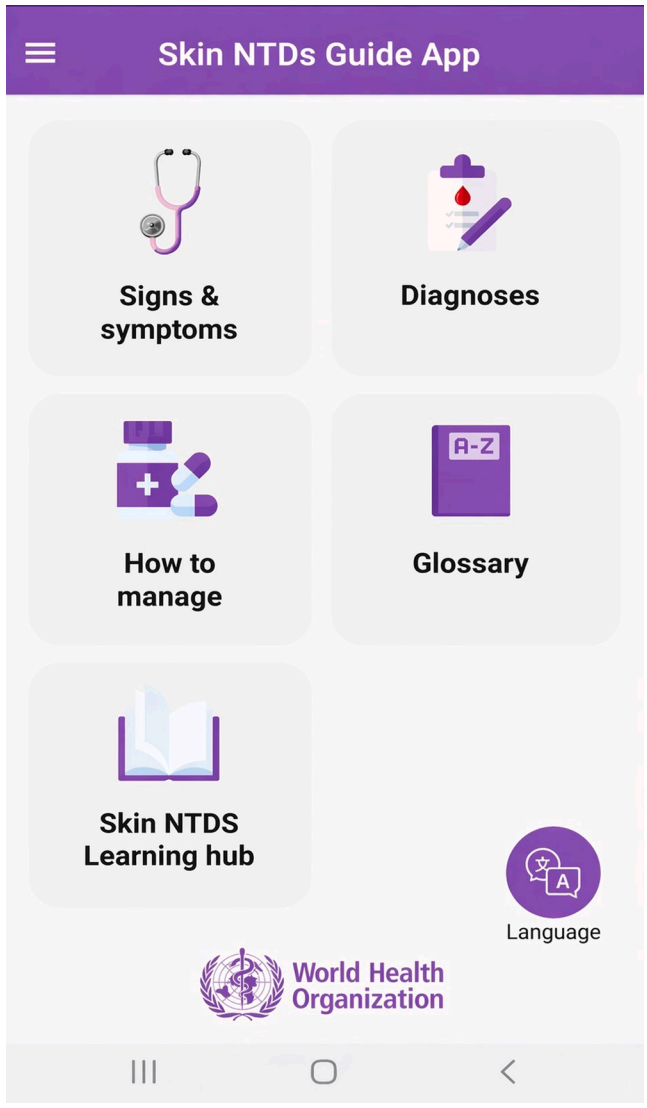



**Figure 1. Screenshot showing the five sections of the WHO Skin NTDs App.**

This version of the WHO Skin NTDs App is publicly available free of charge for Android and iOS and works offline once it is downloaded and installed. Its usability and user experience were assessed in Ghana and Kenya, showing notable usability and user-friendliness (Cano et al, 2024). This indicates that the app could play a crucial role in improving capacity building of FHWs dealing with skin conditions. In order to improve FHWs training, the WHO Skin NTDs App has continued to evolve. In 2024, the app (still in the beta mode and not publicly available in the stores) incorporated a new third CDST using an AI-powered visual classifier (VC). In the next sections, this paper focuses on the VC conceived as a CDST.

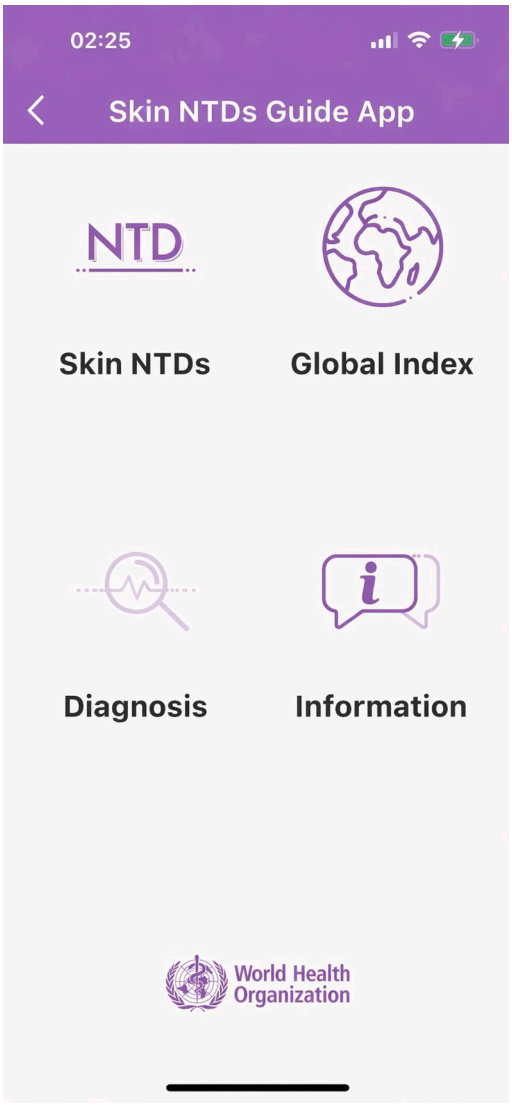



**Figure 2. Screenshot showing the four sub-sections of the Skin NTDs learning hub.**

## RESULTS

Table 1 and Figure 3 show the global performance of the VC. Regarding overall sensitivity, there was a progressive improvement as the value of the number of diseases selected increased (k value), rising from 80.4% for the top-1 to 99.8% for the top-5. In contrast, specificity decreased progressively as the k value increased, ranging from 98.7% for the top-1 to 62.4% for the top-5. The overall accuracy also decreased as the k value increased, reaching 97.7% in the top-1 and 66.66% in the top-5. Precision also had a decreasing trend, from 72.2% to 15% for the top-1 and top-5, respectively. Finally, the overall F1-score, which strikes a balance between precision and sensitivity, also evolved in parallel, achieving values of 75.3% for top-1 and 24% for top-5.

Table 2 shows specific metrics for each disease. The VC identified the target condition as the top-1 diagnosis with a sensitivity of over 75% for 10 out of 12 diseases, whereas it was below 75% for two diseases: chromoblastomycosis was correctly identified as the primary diagnosis in only 20% of cases, while sporotrichosis achieved a sensitivity of 50%.

**Table 1. Global Performance of the Visual Classifier by Sensitivity, Specificity, Accuracy, Precision, and F1-Score, From Top-1 Through Top-5**

| Metric | Top-1 (%) | Top-2 (%) | Top-3 (%) | Top-4 (%) | Top-5 (%) |
|---|---|---|---|---|---|
| Sensitivity | 80.4 | 93.6 | 97.3 | 99.7 | 99.8 |
| Specificity | 98.7 | 90.1 | 80.9 | 71.6 | 62.4 |
| Accuracy | 97.7 | 91.0 | 83.1 | 74.9 | 66.6 |
| Precision | 72.2 | 34.7 | 23.9 | 18.3 | 15.0 |
| F1-Score | 75.3 | 47.8 | 35.6 | 28.4 | 24.0 |

However, when considering the system's top-5 diagnostic suggestions, performance improved, with leprosy reaching a sensitivity of 97.6% and the remaining 11 diseases reaching 100% sensitivity.

Although the specificity was greater than 95% for all diseases at the top-1, it decreased considerably for most of the top-5 diseases. The most significant reductions were seen in Buruli ulcer (from 98.8% to 49.6%), cutaneous leishmaniasis (from 98.8% to 45.8%), and leprosy (from 95.7% to 35.6%).

The precision values in the top-1 were high for most diseases, particularly onchocerciasis (100%), cutaneous leishmaniasis (96.09%), lymphatic filariasis (95%), and post-kala-azar dermal leishmaniasis (87.80%). By contrast, the precision values in the top-1 were low for chromoblastomycosis (20%) and sporotrichosis (25%), reflecting a high proportion of false positives. Low precision values, below 50%, were observed for all diseases in the top-5 and below 5% for chromoblastomycosis (2.94%), mycetoma (2.47%), onchocerciasis (2.4%), and sporotrichosis (1.87%).

A similar decreasing pattern was observed in the F1-score, with values in the top-1 ranging from 20% (chromoblastomycosis) to 100% (onchocerciasis) and a pronounced decrease in the top-5 for all diseases.

Supplementary Table S1 is a spreadsheet showing the values of all the variables used to calculate each metric, that is, true positive, true negative, false positive, and false negative for each disease, from top-1 through top-5. It also contains the calculations for the sensitivity, specificity, accuracy, precision, and F1-score by disease from top-1 through top-5. Additional raw data are available upon request.

## DISCUSSION

This study aimed to develop and evaluate the performance of an artificial intelligence algorithm for detecting skin NTDs. According to our main metric, high sensitivity in the top-5 was observed. This is crucial for the purpose for which this tool was developed: to train FHWs in differential diagnosis. It is important to note that this work develops and evaluates an AI algorithm dedicated to skin NTDs for educational purposes.

### Model performance and preliminary outcomes

When assessing AI systems for dermatology, prioritization and interpretation of metrics depend on the specific dermatological context. For potentially severe conditions like skin cancer, high sensitivity might be crucial to minimizing missed cases. For common, less severe conditions, precision could be emphasized to avoid unnecessary treatments. The

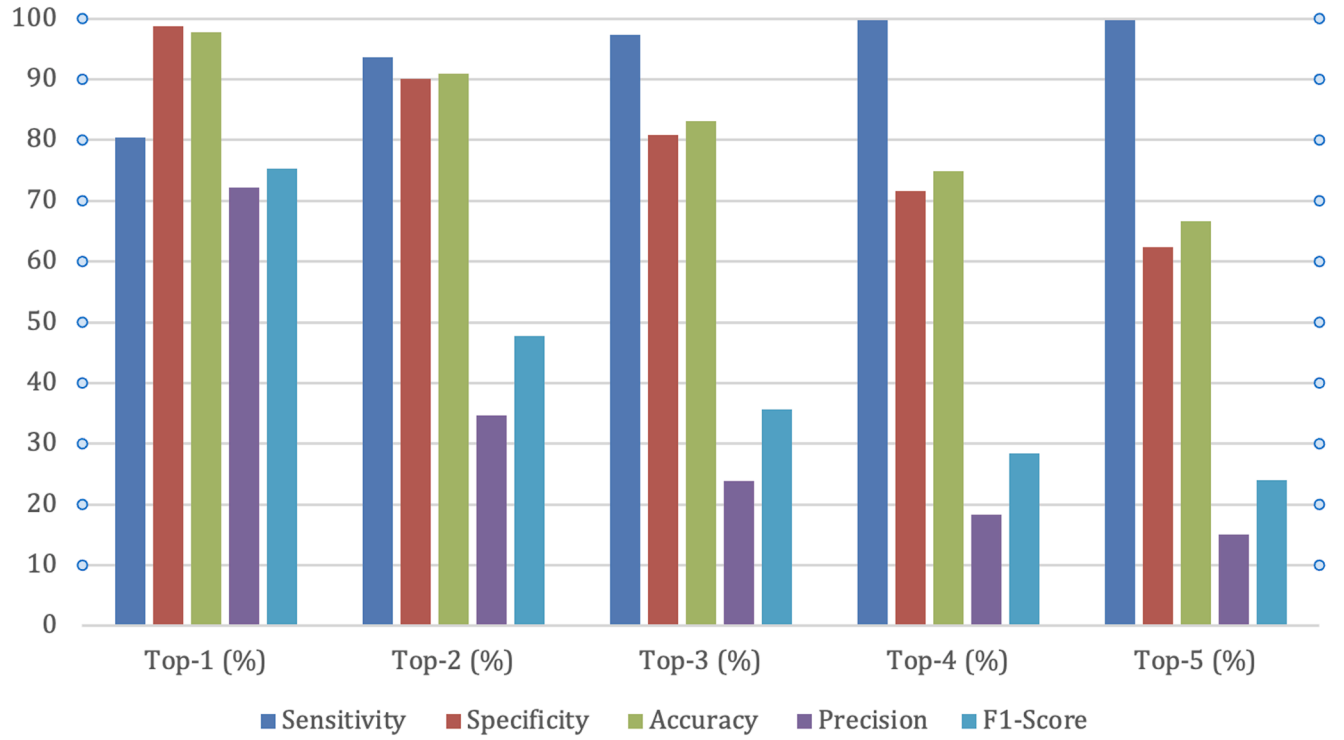



**Figure 3. Global performance of the visual classifier by sensitivity, specificity, accuracy, precision, and F1-Score, from top-1 through top-5.**

WHO Skin NTDs App is conceived as a multiple CDST, not as a diagnostic medical device; that is why, in the VC, the sensitivity of the top-5 differential diagnosis possibilities is the main objective of the tool.

Considering the sensitivity, the results observed with the internal test set can be considered excellent and encourage us to proceed with external and clinical validations in real-world conditions (Esteva et al, 2017; World Health Organization, 2021). Similar sensitivity results were obtained by other authors when evaluating multiclass algorithms (Alruwaili and Mohamed, 2025; Badr et al, 2025).

Focusing on top-5 sensitivity, which better reflects the system's clinical value, the model correctly identified the targeted condition among its top-5 suggestions in 99.8% of cases across the 12 skin NTDs in the dataset. This high sensitivity highlights the VC's potential as a useful CDST. Liu et al (2020) obtained results that were very similar to those of this study. They observed a progressive improvement as the

**Table 2. Per-Disease Evaluation Results. Sensitivity, Specificity, Accuracy, Precision and F1-Score for the Top-1 and Top-5 Differential Diagnoses, by Using the Internal Test Set**

| Class | Test set (n) | % of Images in Test Set | Sensitivity (%) | | Specificity (%) | | Accuracy (%) | | Precision (%) | | F1-score (%) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Top-1 | Top-5 | Top-1 | Top-5 | Top-1 | Top-5 | Top-1 | Top-5 | Top-1 | Top-5 |
| **Buruli ulcer** | 64 | 11.03 | 89.1 | 100 | 98.8 | 49.6 | 97.76 | 55.17 | 90.48 | 19.75 | 89.76 | 32.99 |
| **Chromo.** | 5 | 0.86 | 20.0 | 100 | 99.3 | 71.3 | 80.69 | 71.56 | 20.00 | 2.94 | 20.00 | 5.71 |
| **CL** | 150 | 25.86 | 82.0 | 100 | 98.8 | 45.8 | 94.48 | 55.17 | 96.09 | 39.16 | 88.49 | 56.29 |
| **Leprosy** | 41 | 7.07 | 87.8 | 97.6 | 95.7 | 35.6 | 95.17 | 40.0 | 61.10 | 10.34 | 72.0 | 18.69 |
| **LF** | 11 | 1.90 | 90.9 | 100 | 99.3 | 77.9 | 99.14 | 78.28 | 71.43 | 8.03 | 80.0 | 14.87 |
| **Mycetoma** | 80 | 13.79 | 90.0 | 100 | 98.0 | 51.2 | 96.90 | 57.93 | 87.80 | 2.47 | 88.89 | 39.60 |
| **Oncho.** | 3 | 0.52 | 100 | 100 | 100 | 78.9 | 100 | 78.97 | 100 | 2.40 | 100 | 4.00 |
| **PKDL** | 168 | 28.97 | 90.5 | 100 | 98.1 | 46.6 | 95.86 | 62.07 | 95.00 | 43.30 | 92.68 | 60.43 |
| **Scabies** | 24 | 4.14 | 79.2 | 100 | 98.6 | 53.2 | 97.76 | 55.17 | 70.37 | 8.45 | 74.51 | 15.58 |
| **Sporo.** | 2 | 0.34 | 50 | 100 | 99.5 | 81.8 | 99.31 | 81.90 | 25.00 | 1.87 | 33.33 | 3.67 |
| **Tungiasis** | 4 | 0.69 | 100 | 100 | 99.7 | 88.7 | 99.66 | 88.79 | 66.67 | 5.78 | 80.00 | 10.96 |
| **Yaws** | 28 | 4.83 | 85.7 | 100 | 99.1 | 68.5 | 98.45 | 70.0 | 82.76 | 13.86 | 84.21 | 24.35 |

Abbreviations: Chromo, chromoblastomycosis; CL, cutaneous leishmaniasis; LF, lymphatic filariasis; Oncho, onchocerciasis; PKDL, post-kala-azar dermal leishmaniasis; Sporo, sporotrichosis.

Sensitivity (recall) = TP / (TP + FN); Specificity = TN / (TN + FP).

Accuracy = (TP + TN) / (TP + TN + FP + FN); Precision = TP / (TP + FP).

F1-score = 2 × (Precision × Recall) / Precision + Recall.

number of selected diseases increased. It is important to note that while top-1 and top-2 performance metrics are relevant for rapid triage, our tool was specifically designed to support differential diagnosis by providing five suggestions. This approach reflects its real-world use by FHW in endemic areas, where overlapping clinical presentations and limited diagnostic tools make broader diagnostic support essential. In that sense, similar top-5 reporting strategies have also been adopted in other AI-based dermatology models (Escalé-Besa et al, 2023; Jain et al, 2021).

Although 5 diseases in the WHO NTD photo library, that is, chromoblastomycosis, lymphatic filariasis, onchocerciasis, sporotrichosis, and tungiasis, are underrepresented in the dataset with less than 100 images in the training set each, the VC still achieved a top-5 sensitivity above 95% for all diseases. This shows that, despite limited training samples, the model is capable of including the correct condition among its top predictions, reinforcing its value as an educational tool.

Nonetheless, to ensure consistency and reliability across all metrics and conditions, we recommend temporarily limiting the inclusion of underrepresented diseases in the photo library for the upcoming training and validation iterations of the beta version. The ultimate goal is certainly to reach a minimum image threshold for each disease —ideally, 200 images for conditions with low clinical variability and 500 or more for diseases with diverse morphological presentations. This approach would help ensure more balanced learning and robust evaluation while still allowing the model to support educational use cases where top-5 sensitivity remains high.

The steep drop in specificity in some top-5 results reflect a deliberate design choice: prioritizing sensitivity and inclusiveness to avoid missed cases in the learning process. When reviewing the list of differential diagnoses, FHWs are taught to consider several possibilities before reaching any conclusion. This is critical for highly stigmatizing conditions such as leprosy, where the patient should not be told about the possibility of suffering from that condition unless a solid suspicion is present or, ideally, until confirmation is made. Thus, a lower specificity at the top-5 level must be interpreted as part of a learning-oriented strategy that encourages broader clinical consideration rather than a diagnostic suggestion.

The VC does not suggest referral pathways or provide treatment algorithms. All AI-generated outputs must be considered learning prompts in the form of suggestions. Any decision regarding diagnosis, management, or referral must be made independently by the clinician following established guidelines.

### Strategic impact of AI integration

By embedding an AI-powered CDST based on a VC within the WHO Skin NTDs App, this initiative directly aligns with the WHO's digital health strategy (World Health Organization, 2021) and leverages innovation to strengthen healthcare delivery in underserved regions.

This approach supports: a) capacity building through accessible training tools tailored for FHWs; b) continuous learning, thanks to a dynamic and evolving image classification engine powered by real-world contributions; c) value-based care by promoting early identification and management of NTDs, reducing complications and healthcare burdens; d) stronger patient-FHW relationships by supporting informed decision-making in primary care settings.

### Limitations

Despite the tool's overall promising performance and usability, the data imbalance associated with diseases having a low number of images is a key challenge that should prompt us to interpret cautiously the good results observed for diseases with less than 100 images in the training set. For diseases such as chromoblastomycosis and sporotrichosis, the poor results observed emphasize the need to obtain many more images to retrain and test the VC because it could entail a clinical risk if the clinician relies merely on the tool's performance observed at this moment.

Although we managed to get a stunning 96% of images in the dataset with moderate or dark skin tones, which is fully aligned with the skin tones mainly present in the tropics, we acknowledge, however, that in certain low- or non-endemic areas, or in migrant populations, some patients will have light skin tones and therefore our database may not perform as desired in that context.

Another important limitation is that, as an educational tool, the most appropriate way to evaluate the impact of the app will ultimately be to measure knowledge gain and capacity-building outcomes among frontline health workers, ideally in comparison with standard training modalities, such as lectures or workshops. At this early stage, however, the primary objective was to establish technical feasibility and assess model performance across Skin NTDs. Because the dataset is still evolving and several disease classes remain underrepresented, the model is not yet sufficiently mature to serve as the basis for a rigorous educational impact study.

### Future directions

To guarantee the app's usage in line with its intended purpose, we should ensure that FHWs understand AI-generated insights for educational value as a CDST. For geographical areas with poor Internet connectivity, AI offline capabilities should be envisaged. To diversify and increase image datasets, WHO welcomes partnerships.

To support global implementation, external validation and field-based pilot studies in real-world conditions will be critical, as recommended by WHO (World Health Organization, 2023a). It is out of scope to intend to use this VC solely relying on internal data because that entails unacceptable biases. WHO is currently gathering new images to be used as datasets for external validation. Real-world studies have been done, are ongoing, or are being planned in Cameroon, Colombia, Côte d'Ivoire, India, Indonesia, Kenya, Senegal, and Syria. In conclusion, the WHO Skin NTDs App, now enhanced with AI-driven educational features, could represent a significant advancement in digital health training tools for FHWs responsible for skin NTDs in low- and middle-income countries. It serves as a bridge between traditional training and modern CDST, offering real-time, interactive learning. A prospective study will be conducted to evaluate learning outcomes, diagnostic reasoning improvement, and usability among frontline health workers.

The tool has the potential to become a global model for digital health education and to align with local healthcare needs for its long-term success and global relevance.

## MATERIALS AND METHODS

### Data sources and population

In 2021, we developed a VC algorithm to identify 12 skin NTDs: Buruli ulcer, chromoblastomycosis, cutaneous leishmaniasis, leprosy, lymphatic filariasis (lymphoedema and hydrocele), mycetoma, onchocerciasis, post-kala-azar dermal leishmaniasis, scabies, sporotrichosis, tungiasis, and yaws, excluding noma since the WHO classified it as a skin NTD only in 2023 (World Health Organization, 2023c).

The VC was designed to distinguish between four categories, using an image dataset made up of: skin NTDs (n = 5760), other skin diseases (excluding NTDs; [n = 16,577]), healthy skin (n = 2469), and non-skin images (n = 50,000). Skin NTD images were obtained from WHO publications and institutions collaborating with the WHO Global Initiative on Artificial Intelligence for Skin Conditions (World Health Organization, 2025b). Other skin diseases were obtained from the Fitzpatrick 17k dataset, which include clinical images from DermaAmin and Atlas Dermatologico dermatology atlases that offer diverse coverage of Fitzpatrick skin types (Groh et al, 2022). Healthy skin images were created by cropping healthy areas from skin NTD images to ensure diverse skin tone representation. Finally, non-skin images were randomly selected from the ImageNet dataset (Stanford Vision Lab, 2020).

The training set contained images of different melanin densities: 4% (164/4418) with light skin (equivalent to Fitzpatrick skin types I-II); 26% (1147/4418) with moderate skin (skin types III-IV) and 70% (3107/4418) with dark skin (skin types V-VI) (Table 3).

### Labeling and gold standard

Each image was originally labeled with a diagnostic category for each skin NTD provided by board-certified dermatologists or clinical experts at the time of collection (considered as a Gold Standard).

### Data preprocessing

Prior to training, all images underwent standardized preprocessing to ensure compatibility with the DenseNet-121 architecture. Images were normalized according to the DenseNet-121 CNN architecture specifications, using ImageNet mean and SD values (mean = [0.485, 0.456, 0.406], SD = [0.229, 0.224, 0.225]) for red, green, and blue channels. Additionally, images were cropped to center the lesion area within the frame, ensuring that the region of interest remained the focal point for feature extraction while maintaining consistent input dimensions of 224 × 224 pixels required by the network architecture.

**Table 3. Images Used to Train the AI-Powered Visual Classifier of the WHO Skin NTDs by Disease and Skin Tone**

| WHO Photo library AI | SKIN TONE | | | |
|---|---|---|---|---|
| TRAINING SET | Light | Moderate | Dark | TOTAL |
| Buruli ulcer | 54 | 0 | 435 | 489 |
| Chromoblastomycosis | 5 | 6 | 20 | 31 |
| Cutaneous leishmaniasis | 37 | 728 | 335 | 1100 |
| Leprosy | 0 | 70 | 232 | 302 |
| Lymphatic filariasis | 0 | 0 | 78 | 78 |
| Mycetoma | 1 | 17 | 575 | 593 |
| Onchocerciasis | 0 | 0 | 17 | 17 |
| Post-kala-azar dermal leishmaniasis | 11 | 276 | 1095 | 1382 |
| Scabies | 54 | 35 | 85 | 174 |
| Sporotrichosis | 0 | 15 | 0 | 15 |
| Tungiasis | 2 | 0 | 26 | 28 |
| Yaws | 0 | 0 | 209 | 209 |
| TOTAL | 164 | 1147 | 3107 | 4418 |

Abbreviations: AI, artificial intelligence; WHO, world health organization

### Model architecture

The VC was built upon convolutional neural network (CNN) technology, a type of deep learning neural network architecture frequently used in computer vision tasks. The VC was built upon the CNN - DenseNet 121 architecture and works in two stages. The first stage classifies whether an input image shows signs of a skin NTD or something unrelated, a non-NTD condition, or simply background objects. The CNN processes the image through multiple layers, such as convolutional layers to detect features like edges, textures, and shapes, pooling layers to reduce spatial dimensions, and fully connected layers for classification. The output is a binary classification: either skin NTD (presence of the disease) or non-NTD (no disease or a different condition). If the CNN determines that the image is related to a skin NTD, the system proceeds to the second stage (Figure 4). The second stage focuses on a more detailed classification among the 12 possible skin NTDs. The trained CNN model is used to perform multiclass classification. An inference is made on the model's weights to output the likelihood of belonging to any of the 12 diseases, providing a list of the top five likelihoods of differential diagnosis on the smartphone (Figure 5).

The dataset was initially split into training (75%), validation (15%), and test (10%) sets by using stratified sampling to preserve the class distribution across all subsets.

To prevent data leakage, we implemented several safeguards throughout the modeling process. First, we performed comprehensive duplicate detection using a deep learning-based visual similarity model to identify and remove potential duplicate images across the entire dataset before splitting it.

The validation and test sets remained completely isolated throughout the entire training process, with no information from these sets being used for model development, hyperparameter tuning, or preprocessing parameter selection. This strict separation ensures that model performance metrics reflect true generalization capability rather than overfitting the data seen.

### Evaluation metrics

The CNN is configured to emphasize sensitivity over specificity, allowing for a higher rate of potential false positives rather than false negatives. This approach enables the VC to identify a broad range of possible conditions, which the FHW can then systematically evaluate through clinical history-taking and physical examination. The system functions primarily as an educational tool. A diagnostic medical device is out of scope. The VC's output serves as a learning guide, prompting FHWs to consider various conditions and supporting their clinical reasoning process.

The evaluation of the VC's performance was based on metrics aligned with the model's design goals and intended use. As missing a treatable case is the most significant clinical error in the context of skin NTDs, sensitivity was selected as the primary metric (World Health Organization, 2023a). To account for the intended use of

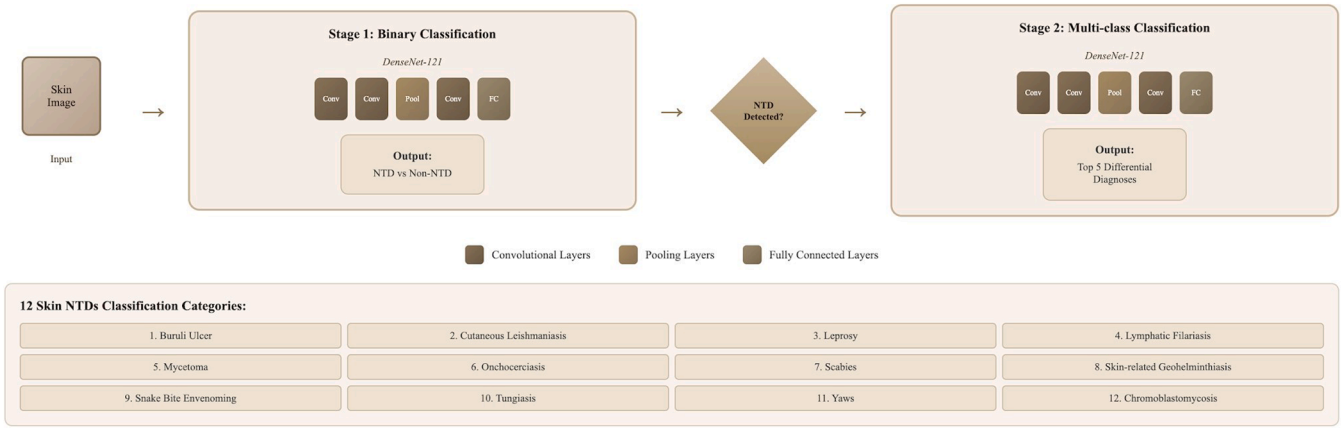



**Figure 4. Two-stage architecture for skin NTDs classification.**

the tool (practitioners review the first four suggestions on the smartphone screen), in addition to classical sensitivity at top-1, top-5 sensitivity was also computed.

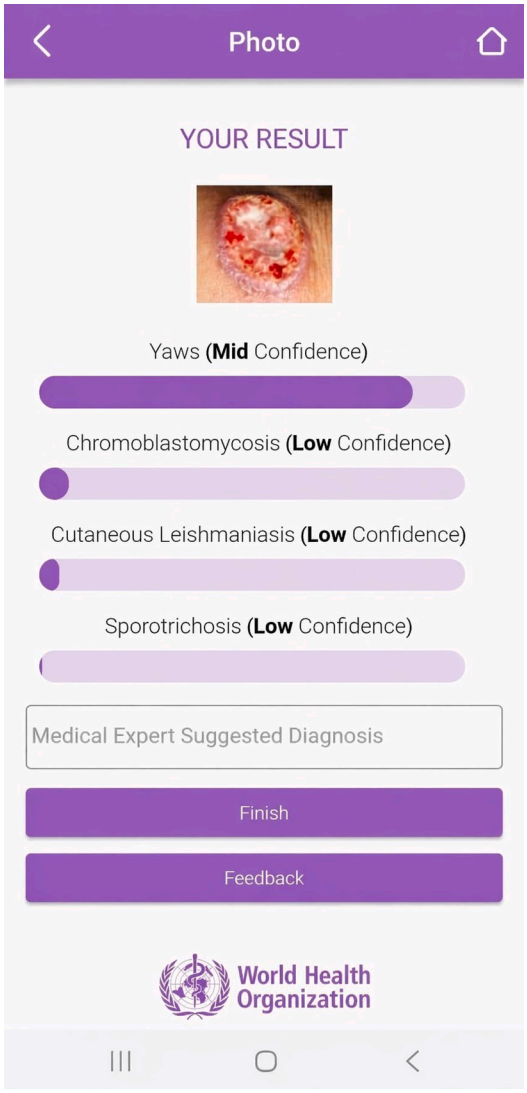



**Figure 5. Screen shot showing an example of the results obtained when an image is analyzed through the visual classifier with artificial intelligence in a smartphone.**

The most common evaluation metrics used for typical supervised machine learning tasks, especially for image identification, including binary, multi-class, and multi-label classification, regression, image segmentation, object detection, and information retrieval, are sensitivity (recall or true positive rate), precision (positive predictive value), specificity (true negative rate), accuracy (overall correctness), and F1-score (a harmonic mean, weighted average of precision and recall).

Per-class sensitivity reveals more about the model than average accuracy, especially if the number of real positive and negative instances is very imbalanced (McGrath, 2021; Rainio et al, 2024). Given the imbalanced nature of our database, the use of accuracy was excluded as a relevant metric. The F1-score provides an overall performance measure.

The k-value metric refers to the number of top predictions considered when calculating top-k sensitivity, specificity, or accuracy metrics (eg, top-1, top-3, and top-5 performance). This is a standard approach in multi-class image classification.

To comprehensively assess the model's performance, metrics were calculated both at a global level using average scores and at a per-class level. This dual evaluation approach allows for the identification of how well the model performs overall and for each individual disease.

An initial internal test was conducted to evaluate the performance of the VC. This test set comprised approximately 10% randomly sampled images from the WHO NTD photo library. Sensitivity was calculated separately for the top-1 and top-5 results, that is, taking into account the first disease shown by the VC or the top-5. Precision, specificity, F1-score, and accuracy were also computed.

### Data protection regulations and training set availability

Each participating institution that signed a license agreement with WHO to share images ensured compliance with data protection regulations and the required consent. Images were anonymized, collected under controlled protocols, and stored in a secure WHO-managed environment under strict data governance rules. No personal identifiable information is retained by the app or the VC. This safeguards patient confidentiality and prevents any unintended secondary use. The NTD image datasets used to train the AI model are not publicly available, as per the will expressed by each participating institution in the license agreement.

### Equitable access to AI tools in low- and middle-income countries

The WHO AI-powered VC is primarily intended to be used in low- and middle-income countries, where dermatology expertise is often scarce and skin NTDs are most prevalent. The app is intentionally designed to be free of charge and compatible with low-connectivity environments.

## ETHICS STATEMENT

Ethical approval was not required for this study. The skin neglected tropical diseases images were obtained from World Health Organization publications and World Health Organization photo gallery with ethical requirements having been previously obtained.

## DATA AVAILABILITY STATEMENT

Data obtained after using the test set of images to assess the performance of the VC are available as supplementary tables. Those tables include the values of the true positive, true negative, false positives, and false negatives for each disease, from top-1 through top-5.

The tables also include the calculations of the five metrics used in this paper, that is, sensitivity, specificity, accuracy, precision, and F1-score, globally and by disease, from top-1 through top-5.

Any other data supporting the findings of this study are available from the corresponding author upon reasonable request.

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## CONFLICT OF INTEREST

The authors state no conflict of interest.

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## AUTHOR CONTRIBUTIONS

Conceptualization: JS, JR, CC, AF, AR, IM; Data curation: AR, IM; Formal analysis: AR, IM, CC, JR, JS, AF, SP, AW, CM; Funding acquisition: JR; Methodology: CC, AF; Project administration and Resources: JL JR JS; Software: FV, MG, JL; Supervision: JR, JS, CC, AF; Validation: AR, IM; Visualization: Writing-original draft: AF; AF, CC, JS, JR; Writing-review & editing: JL, IM, AR, AF, CC, JS, JR, SA, CH, SP, AW, CM

## DECLARATION OF GENERATIVE ARTIFICIAL INTELLIGENCE (AI) OR LARGE LANGUAGE MODELS (LLMs)

The author(s) did not use AI/LLM in any part of the research process and/or manuscript preparation.

## SUPPLEMENTARY MATERIAL

Supplementary material is linked to the online version of the paper at www.jidonline.org, and at 10.1016/j.jid.2026.03.043.

## REFERENCES

Alruwaili M, Mohamed M. An integrated deep learning model with EfficientNet and ResNet for accurate multi-class skin disease classification. Diagnostics (Basel) 2025;15:551.

Badr M, Elkasaby A, Alrahmawy M, El-Metwally S. A multi-model deep learning architecture for diagnosing multi-class skin diseases. J Imaging Inform med 2025;38:1776–95.

Cano M, Ruiz-Postigo JA, Macharia P, Ampem Amoako Y, Odame Phillips R, Kinyeru E, et al. Evaluating the World Health Organization's SkinNTDs app as a training tool for skin neglected tropical diseases in Ghana and Kenya: cross-sectional study. J Med Internet Res 2024;26:e51628.

Chandler DJ, Fuller LC. The skin-A common pathway for integrating diagnosis and management of NTDs. Trop Med Infect Dis 2018;3:101.

Escalé-Besa A, Yélamos O, Vidal-Alaball J, Fuster-Casanovas A, Miró Catalina Q, Börve A, et al. Exploring the potential of artificial intelligence in improving skin lesion diagnosis in primary care. Sci Rep 2023;13:4293.

Esteva A, Kuprel B, Novoa RA, Ko J, Swetter SM, Blau HM, et al. Dermatologist-level classification of skin cancer with deep neural networks [published correction appears in Nature 2017;546:686]. Nature 2017;542:115–8.

Groh M, Harris C, Daneshjou R, Badri O, Koochek A. Towards transparency in dermatology image datasets with skin tone annotations by experts, crowds, and an algorithm. Proc ACM Hum Comput Interact 2022;6:1–26.

Jain A, Way D, Gupta V, Gao Y, de Oliveira Marinho G, Hartford J, et al. Development and assessment of an artificial intelligence-based tool for skin condition diagnosis by primary care physicians and nurse practitioners in Teledermatology practices. JAMA Netw Open 2021;4:e217249.

Liu Y, Jain A, Eng C, Way DH, Lee K, Bui P, et al. A deep learning system for differential diagnosis of skin diseases. Nat Med 2020;26:900–8.

McGrath K. Accuracy and explainability in artificial intelligence: unpacking the terms. In: ICIS 2021 proceedings. TX: AIS on behalf of ICIS; 2021. p. 1–9.

Mwageni N, van Wijk R, Daba F, Mamo E, Debelo K, Jansen B, et al. The NLR SkinApp: testing a supporting mHealth tool for frontline health workers performing skin screening in Ethiopia and Tanzania. Trop Med Infect Dis 2024;9:18.

Rainio O, Teuho J, Klén R. Evaluation metrics and statistical tests for machine learning [published correction appears in Sci Rep 2024;14:15724] Sci Rep 2024;14:6086.

Ruiz-Postigo JA, Pathak P, Asiedu K. Skin health for all: update on skin neglected tropical diseases with a focus on Buruli ulcer and yaws. Wkly Epidemiol Rec 2025;100:239–50.

Stanford Vision Lab. ImageNet [online database]. https://image-net.org/index.php; 2020 (accessed November 13, 2024).

Thunga G, Khan S, Poojari PG, Rajan AK, Rashid M, Kaur H, et al. Artificial intelligence in neglected tropical diseases. Manipal: Academic Press; 2024.

Venkatesh KP, Raza MM, Nickel G, Wang S, Kvedar JC. Deep learning models across the range of skin disease. NPJ Digit Med 2024;7:32.

World Health Organization. Recognizing neglected tropical diseases through changes on the skin: a training guide for front-line health workers. https://www.who.int/publications/i/item/9789241513531; 2018 (accessed November 13, 2024).

World Health Organization. Global strategy on digital health 2020-2025. https://www.who.int/docs/default-source/documents/gs4dhdaa2a9f352b0445bafbc79ca799dce4d.pdf; 2021 (accessed November 13, 2024).

World Health Organization. Regulatory considerations on artificial intelligence for health. https://www.who.int/publications/i/item/9789240078871; 2023a (accessed November 13, 2024).

World Health Organization. Skin-related neglected tropical diseases: WHO launches new version of the WHO Skin NTDs mobile application. https://www.who.int/news/item/09-10-2023-skin-related-neglected-tropical-diseases–who-launches-new-version-of-the-who-skin-ntds-mobile-application; 2023 (accessed July 31, 2025).

World Health Organization. WHO officially recognizes noma as a neglected tropical disease. https://www.who.int/news/item/15-12-2023-who-officially-recognizes-noma-as-a-neglected-tropical-disease; 2023 (accessed July 31, 2025).

World Health Organization. Neglected tropical diseases. https://www.who.int/health-topics/neglected-tropical-diseases#tab=tab_1; 2025 (accessed July 31, 2025).

Yotsu RR, Kouadio K, Vagamon B, N'guessan K, Akpa AJ, Yao A, et al. Skin disease prevalence study in schoolchildren in rural Côte d'Ivoire: implications for integration of neglected skin diseases (skin NTDs). PLoS Negl Trop Dis 2018;12:e0006489.

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